Project 3 - Milestone 2
Detecting Credit Card Fraud Using Machine Learning

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Introduction

Credit card fraud is a serious issue that affects financial institutions and customers around the world. As online payments increase, detecting fraudulent transactions becomes more important. Machine learning techniques can help identify suspicious activity by analyzing patterns in large datasets.

Business Problem

Financial institutions process millions of transactions every day, making manual fraud detection impossible. The main goal of this project is to build a machine learning model that can identify fraudulent transactions while minimizing false alarms for legitimate users.

Dataset

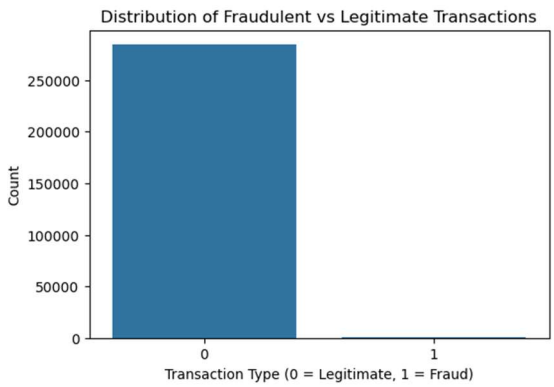
This project uses the Credit Card Fraud Detection dataset available on Kaggle. The dataset contains 284,807 transactions, including 492 fraudulent cases. Most variables were transformed using Principal Component Analysis (PCA) to protect sensitive information. The dataset includes transaction time, amount, and a class variable indicating fraud or legitimate transactions.

Methods

Exploratory data analysis was used to understand the structure of the dataset. Because the dataset is highly imbalanced, techniques such as SMOTE were used to improve the representation of fraudulent transactions. Machine learning models such as Logistic Regression, Random Forest, and XGBoost were considered for fraud detection. Model performance was evaluated using precision, recall, F1-score, and precision–recall curves.

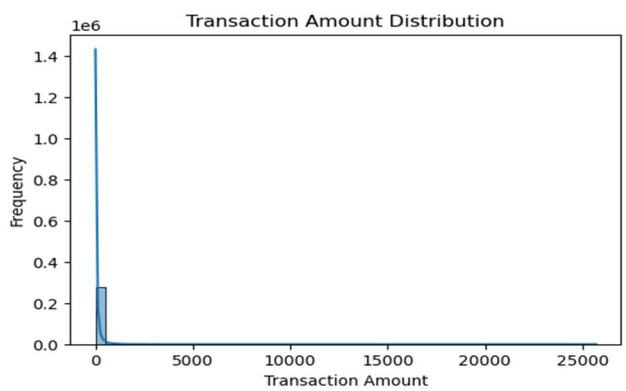
Illustrations

Figure 1. Distribution of Fraudulent vs Legitimate Transactions



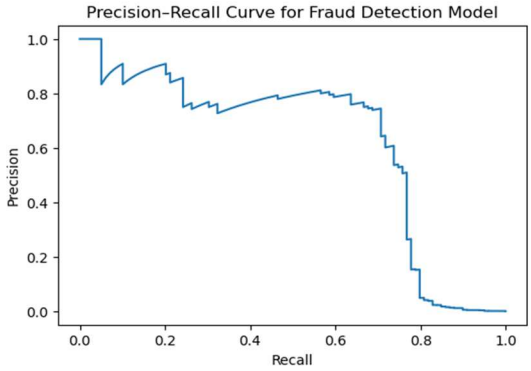
This plot shows the number of legitimate and fraudulent transactions in the dataset. Most transactions are legitimate, while only a small number are fraud. This shows that the dataset is highly imbalanced. Because of this imbalance, special techniques may be needed when building the fraud detection model.

Figure 2. Transaction Amount Distribution



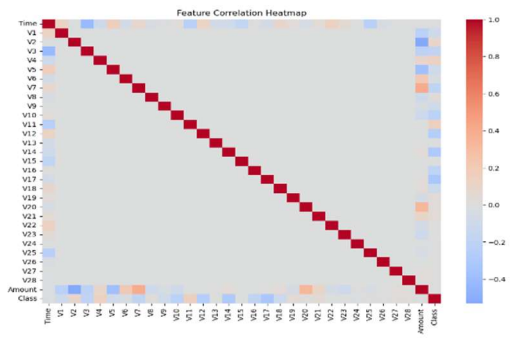
This plot shows how transaction amounts are distributed. Most transactions involve small amounts, while large transactions are rare. The distribution is highly skewed to the right. This means that the majority of purchases are low-value transactions.

Figure 4. Precision–Recall Curve for Fraud Detection Model



This curve evaluates the performance of the fraud detection model. The model starts with high precision when recall is low. As recall increases, precision decreases. This shows the trade-off between detecting more fraud cases and increasing false positives.

Figure 4. Feature Correlation Heatmap



This heatmap shows the correlation between the features in the dataset. Most variables have very weak correlations with each other. This is expected because the variables were created using PCA. The fraud class also has low correlation with individual features, which means fraud detection requires analyzing multiple variables together.

Key Findings

The visualizations show that fraudulent transactions represent a very small percentage of the dataset. Transaction amounts are mostly small and follow a skewed distribution. The precision–recall curve demonstrates the trade-off between identifying fraudulent transactions and avoiding false positives.

Ethical Considerations

Fraud detection systems must protect customer privacy and ensure responsible use of financial data. Another concern is the impact of false positives, where legitimate transactions are incorrectly flagged

as fraudulent. Models must be designed carefully to minimize inconvenience to customers.

Conclusion

Machine learning provides a powerful approach for detecting fraudulent transactions in financial datasets. Although the dataset is highly imbalanced, techniques such as SMOTE and appropriate evaluation metrics can improve model performance. Fraud detection systems can help financial institutions reduce losses and improve transaction security.

Audience Questions with the Answers:

Question 1: Why is fraud detection difficult in this dataset?

Answer: Fraud detection is difficult because fraudulent transactions represent less than 1% of the data.

Question 2: Why are precision and recall important metrics?

Answer: They measure how well the model detects fraud while minimizing incorrect predictions.

Question 3: Why is the dataset imbalanced?

Answer: In real financial systems, fraudulent transactions are rare compared to legitimate transactions.

Question 4: What is SMOTE?

Answer: SMOTE is a technique that creates synthetic examples of the minority class to balance the dataset.

Question 5: Why were PCA features used in the dataset?

Answer: PCA was used to protect sensitive customer information while preserving patterns in the data.

Question 6: What challenges did the project face?

Answer: The biggest challenge was the extreme imbalance between fraudulent and legitimate transactions.

Question 7: Why use machine learning for fraud detection

Answer: Machine learning can analyze large amounts of data and detect patterns that are difficult for humans to identify.

Question 8: What does the precision–recall curve show?

Answer: It shows the relationship between detecting more fraud cases and increasing false positives.

Question 9: What ethical issues should be considered?

Answer: Protecting customer privacy and minimizing false fraud alerts are important ethical concerns.

Question 10: How can this model be improved in the future?

Answer: Future work could include deep learning models and additional transaction data to improve detection accuracy.

References

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